give false positives. Hand investigation does not reveal cancer cells in very early stages when they may lie deep inside the breast and the tumour is not as large.

Last month, Datar Cancer Genetics launched EasyCheck-Breast in India, which allows breast cancer to be diagnosed with a simple blood test available for around Rs 9,000. The test analyses the presence of circulating tumour cells in a person's peripheral blood. EasyCheck-Breast is clinically proven to have above 92 per cent sensitivity (the ability to detect cancer in patients), and above 99 per cent specificity (the ability to differentiate cancer patients from healthy individuals). You have cancer if your blood sample throws up a certain gene, but its absence does not necessarily mean you don't have it. In other words, the test does not throw up any false positives, even if it is prone to false negatives.

Datar Cancer Genetics also offers EasyCheck 360, which helps in the early detection of 30 common cancers with high accuracy, and Trublood, which analyses tumour cells in blood samples to highlight the relevant biomarkers that enable specific treatment for the cancer (much like an invasive biopsy). Other companies that offer molecular diagnostics in India include Myriad Genetics, Sayre Therapeutics and Karkinos Healthcare. For high-risk patients who can afford the current price, doctors say it is a ray of hope. "The cure to cancer is going to be at the cellular level—both for detecting it early by figuring out the genetic defects and treating it by correcting the defect. The human DNA is so long that it will take time to figure it all out—but we are certainly on the right track," says Dr Sarin.

KNOW YOUR ENEMY

Affordability and accessibility will continue to be the two main challenges in cancer care, especially where economic disparities are as wide as they are in India. Not only are many of the new treatments and diagnostic tools still in the early stages, hey also remain extremely expensive. "The good news," says Dr Kumar, "is that these solutions have reached advanced clinical stages. The next step will be to get the costs to come down."

Affordability, experts explain, is predicated on upstream challenges: developing immuno-oncological or molecular oncological products is expensive. It took Purwar nearly four years just to understand how to master the process of developing a CAR-T product. ImmunoACT had to raise around $6.2 million (Rs 4.9 crore) in funding just to get started on CAR-T product development.

To reduce costs would be to reduce quality, says Dr Murthy. "Quality is non-negotiable. It may add to a technology's cost and time, but it is necessary and justifiable," she says. From sample collection to analysis in the lab, companies say that the new technologies need every bit of quality control possible, as the smallest error can result in serious consequences.

Dr Devi Shetty, chairman and executive director of Narayana Health, who has set up the country's first hospital-based CAR-T unit in Bengaluru, says they have transplanted cells in around nine patients, with encouraging results. But the unit itself took three years to be set up. And while it is doing well in terms of outcomes, the costs are too high (since the



RAGUL KRISHNAN

HOPE FLOATS
Serum Institute's vaccine manufacturing unit in Pune

"Anything in healthcare is always expensive at first. Gradually, with changes in technology and economies of scale, costs start coming down"



DEVI SHETTY
Chairman & ED, Narayana Health